

SYMMETRY VERSUS ASYMMETRY IN SEXUAL ISOLATION EXPERIMENTS

JOHN M. RINGO, HAROLD B. DOWSE, AND STEVEN LAGASSE
Department of Zoology, University of Maine, Orono, ME 04469

Abstract.—Two hypotheses predicting the ancestral or derived status of populations and based on asymmetrical mate discrimination (Kaneshiro, 1976; Watanabe and Kawanishi, 1979) were tested using nine laboratory populations of *D. simulans*, a highly outcrossed ancestral population, and eight populations derived from it via founder-flush-crash cycles. The data from individual mating tests using pairwise combinations of these populations fit the Kaneshiro hypothesis reasonably well, rejecting the Watanabe-Kawanishi hypothesis. However, more powerful tests rejected the Kaneshiro hypothesis for the data we analyzed. The values for derived females predicted by the Kaneshiro hypothesis were biased: they were consistently high for derived males and consistently low for ancestral males. We propose a hypothesis, based on variation in mating propensities and symmetrical mate discrimination. We assessed the power of Kaneshiro's and our hypotheses to predict the number of matings between derived females and derived males by plotting predicted vs. observed values and fitting these points to the expected line of unit slope passing through the origin. Predictions of our hypothesis explained more of the variance ($r^2 = 0.87$) than predictions of the Kaneshiro model ($r^2 = 0.63$). While asymmetrical sexual isolation undoubtedly occurs between some species, its existence cannot be determined simply by measuring mating frequencies in a single experiment.

Received July 18, 1985. Accepted April 22, 1986

Sexual isolation is an important barrier restricting gene flow between species (Mayr, 1963). Much of the work on sexual isolation has, for practical reasons, utilized species of *Drosophila* (Spieth and Ringo, 1983), but broad generalizations from this work seem to apply widely to sexually reproducing animals (Mayr, 1963). In *Drosophila*, sexual isolation is often incomplete between closely related species and nearly always complete between distantly related species (Spieth, 1974). Thus, information about sexual isolation is helpful in phylogenetic analysis (Patterson and Stone, 1952; Spieth, 1974). The behavioral causes of sexual isolation are of extreme evolutionary as well as ethological interest (Spieth, 1968).

Even in early experiments measuring sexual isolation, there were asymmetries, and it was apparent that these asymmetries could be caused by unequal mating propensities (e.g., Stalker, 1942; Dobzhansky, 1944; Dobzhansky and Streisinger, 1944; Patterson et al., 1947). Bateman (1949) specifically addressed and solved this problem when he introduced a "joint isolation index." Levene (1949), too, drew attention to apparent asymmetries in sexual isolation which could most simply be explained in terms of differences in mating propensity. Experimental verification of interspecific

and strain differences in mating propensity was soon to follow (Spieth, 1949, 1951), including a quantitative assessment of the contribution of male mating propensity to sexual isolation (Spieth, 1951). Little experimental work has been done to follow up Spieth's original observations; rather, mating propensity either has been considered a "nuisance" factor in determinations of sexual isolation (e.g., Petit et al., 1976) or else has been ignored (cf. Barton and Charlesworth, 1984).

Attention has been refocused on asymmetries observed in experiments measuring sexual isolation, now with the idea that these asymmetries will have predictive value in phylogenetic analysis (reviewed by Kaneshiro, 1983; Giddings and Templeton, 1983). Kaneshiro (1976) observed asymmetry in four of six determinations of sexual isolation between pairs of Hawaiian *Drosophila* whose phylogenetic relationships ("ancestral" vs. "derived") were inferred from biogeographic, cytogenetic, and morphological considerations: ancestral males mated with ancestral and derived females with approximately equal frequency, whereas derived males showed strong homogamy (mated intraspecifically much more than interspecifically). Based on these experiments, but without further data analy-

sis, Kaneshiro stated that there was a second asymmetry: ancestral females mated with strong homogamy, whereas derived females mated randomly with ancestral and derived males. He then hypothesized that this second asymmetry can be used to predict the ancestral or derived status of two populations or closely related species if two conditions are met: 1) the (allopatric) populations are related by founder events, and 2) the ancestral population has not been subjected to drastic bottlenecks in population size subsequent to their divergence, but the derived population has (Kaneshiro, 1976; Giddings and Templeton, 1983). Watanabe and Kawanishi (1979) suggested an alternative hypothesis based on asymmetries in sexual isolation experiments: females of the derived species mate homogamically to a greater extent than females of ancestral species. They analyzed only differences between species in male tendencies to mate homogamically, just as Kaneshiro (1976) did. Both hypotheses have been tested experimentally, some workers favoring one hypothesis, some favoring the other hypothesis, and some being skeptical of both (reviewed by Giddings and Templeton, 1983). D. G. Gilbert (pers. comm.) has performed extensive quantitative analyses which generally support the Kaneshiro hypothesis. Barton and Charlesworth (1984) pointed out that many of the recently published examples of asymmetrical isolation could be most simply explained by reduced mating propensity in the derived populations.

One way to test the competing hypotheses in a laboratory experiment is to establish an ancestral population, to derive several "allopatric" subpopulations from it via founder-flush-crash cycles, and to measure rates of mating within and between the ancestral and the derived populations. We are aware of two experiments which meet these criteria, those of Powell (1978; Dodd and Powell, 1985) and Ringo et al. (1985). Kaneshiro (1983) reported that Powell's data fit the Kaneshiro hypothesis reasonably well, and better than they fit the Watanabe-Kawanishi hypothesis, which we confirm here using a different analysis.

In this paper, we formalize these hypotheses and test the asymmetry hypotheses us-

ing the data of Ringo et al. (1985). In addition, we propose and test a quantitative, predictive hypothesis which assumes symmetry in assortative mating, i.e., no interaction between homogamy and either males or females. Since our hypothesis, as well as those of Kaneshiro and of Watanabe and Kawanishi, are directly related to laboratory measurements of mating frequencies, a hypothetical mating experiment (*gedankenexperiment*) is described in the next section as an introduction to the hypotheses.

Hypotheses of Mating Frequencies

A Gedankenexperiment.—Consider a "multiple choice" mating experiment using two types of animal (A and B); the object of the experiment is to measure the frequencies of the four kinds of matings, A females with A males, A with B, B with A, and B with B. In the experiment, N females and N males of each type, all sexually mature virgins of the same age, are placed together under constant conditions appropriate to the species. Matings of each kind are observed and scored for a fixed duration of time. The duration is chosen so that about half the animals mate over that time. If the two types are not morphologically distinct, they are marked in a way that does not interfere with mating frequencies. We replicate the experiment until a large number of matings have been observed and scored. Other experimental designs (Spieth and Ringo, 1983) are frequently encountered. Little is known about the biological implications of choosing one design over another, but the results take the same form (below) and the analysis is, for our purposes here, the same so long as a balanced design (equal numbers of potential matings of each type) is used. The results of the experiment are to be compared with those expected under some probabilistic model, whose terms are defined in Table 1.

In the left-hand part of the table, the four entries, P_{ij} , are probabilities that sum to 1; e.g., the probability that a mating will occur between a type A female and a type A male is P_{11} . In a completely random model the cell entries are all equal, $P_{ij} = 0.25$.

The row and column marginals are sums; e.g., the probability that a mating will be accomplished by a type A male is $P_{1.} = P_{11}$

TABLE 1. Probabilities and observed values in the *gedankenexperiment*.

Probabilities					Observed Values				
Female					Female				
Male		A	B	Total	Male		A	B	Total
	A	P_{11}	P_{12}	$P_{1.}$		A	x_{11}	x_{12}	$x_{1.}$
	B	P_{21}	P_{22}	$P_{2.}$		B	x_{21}	x_{22}	$x_{2.}$
	Total	$P_{.1}$	$P_{.2}$	1		Total	$x_{.1}$	$x_{.2}$	T

+ P_{12} , and the probability that a mating will be accomplished by a type B male is $P_{2.} = P_{21} + P_{22} = 1 - P_{1.}$. The row marginals represent relative male mating propensities, and the column marginals represent relative female mating propensities (Spieth and Ringo, 1983). Variation in male mating propensity has often been observed in *Drosophila* and has been studied ethologically and genetically (reviewed by Spiess, 1970). Variation in female mating propensity, while it is important as a cause of sexual selection, has received less attention (Spieth and Ringo, 1983).

If mating is not assortative, then the P_{ij} values can be predicted from the marginals; e.g., $P_{11} = (P_{1.})(P_{.1})$. Assortative mating is an interaction between female and male types not predictable from the marginal probabilities. An excellent measure of assortative mating is Schaffer's (1968) discrimination index, D , but it is not conveniently calculated by hand, since it is determined by an iterative procedure. A good alternative measure of assortative mating, derived from that of Levene (Ehrman and Petit, 1968), is $Y_A = (\sqrt{a} - 1)/(\sqrt{a} + 1)$ where $a = P_{11}P_{22}/P_{12}P_{21}$, the cross product ratio (Spieth and Ringo, 1983); whenever $a > 1$ and $Y_A > 0$, assortative mating is positive, and whenever $a < 1$ and $Y_A < 0$, assortative mating is negative. This alternative measure fails if any cell probability is zero, a problem which can be solved by smoothing the data (Spieth and Ringo, 1983). Although the sampling scheme of mating experiments may be hypergeometric, and although the cross-product ratio for small sample sizes is biased (Goux and Anxolabehere, 1981), this is a minor matter if $\log(a)$ is used (S. Fienberg, pers. comm.), which is the case for the test of significance of a recommended by Bishop et al. (1975).

It is tempting to use the probability of homogamic matings ($A \times A$ and $B \times B$), $P_{11} + P_{22}$, as a measure of sexual isolation. For convenience we call the situation in which $P_{11} + P_{22} > 0.5$ "homogamy." Homogamy is highly correlated with positive assortative mating but can be influenced by unequal mating propensity.

The right-hand table of observed values (Table 1) is structured in the same way as the left-hand table of probabilities; e.g., x_{11} matings occurred between type A females and type A males; $x_{2.}$ matings involved type B females; $x_{.1}$ matings involved type A males; and there was a total of T matings. The numbers of matings expected under the model on the left are simply $(P_{ij})(T)$; e.g., one expects $(P_{11})(T)$ matings between type A females and type A males. The goodness-of-fit of the model to the observations can be assessed by G^2 or X^2 (Sokal and Rohlf, 1981). An experiment of this type yields only crude estimates of behavioral probabilities such as mating propensity, because the frequencies of matings after a fixed interval of time are not the same as instantaneous probabilities of mating (Taylor, 1975; Ringo et al., 1986b). We ignore this problem in the present paper.

Turning from the general to the specific, imagine an experiment in which $N = 5$ animals of each sex-type combination are used in each of 40 replications, with the results shown in Table 2.

The right-hand table was obtained by dividing each entry and marginal of the left-hand table by the total of 200; thus, the table of frequencies resembles the table of probabilities in the model discussed above. To distinguish observed frequencies from probabilities, we refer to the entries in the right-hand table as f_{ij} , the row totals as $f_{i.}$, and the column totals as $f_{.j}$. An obvious but

TABLE 2. Results from a hypothetical experiment in which five animals of each sex-type combination are used in each of 40 replications.

Numbers Mating					Mating Frequencies				
Female					Female				
		A	B	Total			A	B	Total
Male	A	100	40	140	Male	A	0.5	0.2	0.7
	B	20	40	60		B	0.1	0.2	0.3
	Total	120	80	200		Total	0.6	0.4	1.0

important datum of the *gedankenexperiment* is the total number of matings, $T = 200$; the overall proportion of individuals mating was 200 females/400 females = 0.5. In addition, there are five important deviations from a completely random model, easily confirmed by Chi-square tests. 1) Type A males copulated more than type B males ($X^2 = 32$). The relative mating propensity of type A males was 0.7 and that of type B males was 0.3. 2) Type A females copulated more than type B females ($X^2 = 8$). The relative mating propensity of type A females was 0.6 and that of type B females was 0.4. 3) There was positive assortative mating (Schaffer's $D = 28.6$, $X^2 = 23.83$; $a = 5$), because more homogamic matings occurred than could be predicted from the mating propensities (marginal mating rates). 4) There were more homogamic than heterogamic matings ($X^2 = 32$). 5) There was an interaction between homogamy and type of female: type A females mated homogamically ($X^2 = 53.33$), whereas type B females did not ($X^2 = 0$). It is not appropriate in a real example to perform five Chi-square tests on a 2×2 table; the purpose here is to point out that any one of the deviations from random behavior can be very simply tested for significance. There are six deviations from randomness in a 2×2 table of usual interest (row, column, and diagonal effects; interaction between row and column, diagonal and row, and diagonal and column); one may choose no more than three to test statistically from one table.

There are several behavioral explanations for the deviations from random mating in the *gedankenexperiment*, all equally plausible a priori. One often-used explanation is that interactions among the four types of animals (e.g., female discrimination be-

tween males) underlie all the variation. For example, f_{11} could be greater than f_{21} purely because A females accept A males more readily than B males, or else the difference could entirely be caused by a reluctance of B males to court A females. What about the differences in mating propensity? They could be trivial outcomes with no biological meaning; e.g., type A males and type B males might have courted with equal latency, persistence, and intensity, but the high discrimination of type A females could completely explain the difference in relative mating success of A and B males. A second possibility is that the mating frequencies vary in part because of mating propensities (vigor, persistence, and intensity of male courtship, and acceptance threshold of females), and in part because of interactions as discussed in the previous explanation. If the second explanation were correct, mating propensities would tend to be stable from experiment to experiment; e.g., type A females would tend to mate more rapidly than type B females, for a wide range of male types. This explanation gains credence from the fact that different populations and laboratory strains often mate at consistently high or low rates (Spiess, 1970; Spieth and Ringo, 1983).

In addition to being in accord with known behavioral attributes of *Drosophila* and other complex animals (Bastock, 1967), the second explanation entails a predictive hypothesis. All the variation in a mating experiment, including female-homogamy interaction, is explained by three parameters: the relative mating propensity of either type of female (f_j), the relative mating propensity of either type of male (f_i), and the degree of assortative mating, a function of $a = f_{11}f_{22}/f_{12}f_{21}$. Conversely, if one knows f_i , f_j ,

a , and T , one can calculate without error the numbers of matings of each type, x_{ij} . We contend that the three parameters have a real behavioral basis. We propose a hypothesis using empirical estimates of P_i , P_j , and a , and test its predictive power.

The Kaneshiro Hypothesis.—In Kaneshiro's hypothesis (Kaneshiro, 1976; Kaneshiro, 1983; Giddings and Templeton, 1983), qualitative and quantitative predictions about relative rates of mating are made for ancestral and derived populations or species. The hypothesis assumes the following scenario. A derived population arises by founder events from an ancestral population. Subsequent to the split, there is little or no gene flow between the two populations; the ancestral population is not subjected to drastic population bottlenecks, but the derived population may be. The derived males lose courtship elements from their repertory, and there is an evolutionary adjustment to this loss, after which derived females tend not to discriminate courting males on the basis of the lost courtship elements. Ancestral males retain all their courtship elements, and ancestral females correspondingly discriminate among males using the entire ancestral courtship ritual. In mating experiments, ancestral females mate at higher rates with ancestral males than with the courtship-deficient derived males, whereas derived females mate at equal rates with ancestral and derived males, because they do not discriminate against males on the basis of the lost elements. Under these assumptions, $P_{11} > P_{21}$ and $P_{12} = P_{22} = P_2/2$; the hypothesis does not address the issue of differences in mating propensity between ancestral and derived types; i.e., it tacitly entails the assumption that mating propensities are equal. The hypothesis does not predict P_{11} or P_{21} , nor does it predict the P_{ij} 's from information extrinsic to the experiment.

The Watanabe-Kawanishi Hypothesis.—This hypothesis (Watanabe and Kawanishi, 1979; Watanabe and Kawanishi, 1981) assumes that for any two closely related species, one is ancestral and the other is derived. Watanabe and Kawanishi suggested that an incipient derived species would have a small population size relative to its ancestral species and that the potential for gene flow between the two would exist. Un-

der those conditions, there would be strong selection in the derived species for sexual isolation between derived females and ancestral males, because derived females would encounter ancestral males at a high rate. Analogous selection in the ancestral species would be much weaker. Under these assumptions, derived females will mate homogamically to a greater extent than ancestral females, i.e., $P_{22}/P_{12} > P_{11}/P_{21}$, but none of the P_{ij} 's are predicted.

An Ethological Hypothesis of Mating Behavior.—Our hypothesis relates known attributes of mating behavior to mating frequencies, measured in an experiment such as the *gedankenexperiment*, above (again, two types, A and B, are considered). No assumptions are made about the ancestral or derived status of the two types. Mating frequencies are predicted under the hypothesis by three parameters: female mating propensity, male mating propensity, and assortative mating. The mating propensity of females reflects a readiness to mate. In the classic ethological model of Bastock and Manning (1955), females have an acceptance threshold such that they allow or solicit copulation when the cumulative stimulation of male courtship exceeds this threshold. This readiness to mate, which is influenced by many factors, such as age and genotype (reviewed by Spieth and Ringo, 1983), can be quantified by the mean frequency of mating, averaged over experiments in which a wide range of male types is used. The mating propensity of males reflects the frequency, intensity, and quality of courtship, all of which vary among males (Spieth and Ringo, 1983). Assortative mating is assumed to be the net result of eight potential affinities or discriminatory tendencies: affinities that A males have for A and for B females, those that A females have for A and B males, and similarly for B females and B males. Any of these affinities may or may not exist for a given pair of species or populations. Within species, males usually appear to court indiscriminately with respect to female phenotype (Spieth and Ringo, 1983), whereas between species, males often show strong affinities (Spieth, 1951; Schilcher and Dow, 1977; Wood and Ringo, 1980). Whenever males court intraspecifically more often than interspecific-

ically, assortative mating and homogamy tends to result. Females respond to the quality as well as the intensity and quantity of male courtship (Spieth and Ringo, 1983), and show corresponding affinities for phenotypic variation in males. Despite this, females of some species appear to show little or no more affinity for conspecific males than for males of another closely related species (Wood and Ringo, 1980). The pattern of affinities between two particular types may be precisely that specified in either Kaneshiro's or Watanabe and Kawanishi's hypothesis. For simplicity, in this paper the effect of these affinities on the probabilities of mating (P_{ij}) are assumed to be symmetrical, since we suppose that the pattern of affinities varies from case to case, and since this pattern is extremely difficult to determine experimentally. Within species, mating propensities usually have a larger potential than assortative mating for influencing mating probabilities; between species, the opposite is true. The probabilities of mating combinations are then:

$$P_{ij} = f_i f_j + k, \text{ for } i = j,$$

and

$$P_{ij} = f_i f_j - k, \text{ for } i \neq j,$$

where f_i and f_j are the mating frequencies of type i males and type j females, respectively, determined experimentally and normalized so that $f_1 + f_2 = f_{.1} + f_{.2} = 1$. The homogamy factor, k ($0 < k < 1$), is found by solving the quadratic equation,

$$(a - 1)k^2 - (af_1.f_2 + af_2.f_1 + f_1.f_1 + f_2.f_2)k + (a - 1)f_1.f_2.f_1.f_2 = 0,$$

where a is the cross-product ratio from the appropriate experiment (see Materials and Methods). An example in which the P_{ij} values are estimated is given in the Appendix. Estimates of P_i and P_j tend to be more accurate than estimates of k . For some strains, mating propensities are so stable that the numbers of matings, x_{ij} , can be estimated without knowing T , the total number of matings; i.e., for such strains, T can be estimated fairly accurately from the raw mating frequencies in independent tests and the number of animals used in the experiment.

MATERIALS AND METHODS

Experimental Lines.—The data used here were collected in an experiment testing two hypotheses of speciation (Ringo et al., 1985), in mating tests among nine populations. One ancestral base population (BASE) was established by pooling many wild-caught stocks of *Drosophila simulans* from recent worldwide field collections. During the three-year experiment, BASE was maintained in a population cage at an average size of 8,000 adults. Eight drift populations were founded by single fertile pairs chosen at random from BASE. Drift populations were treated the same as BASE except that every six months, the size of each drift population was reduced to 2; i.e., single fertile pairs were chosen randomly from each drift population. Thus, each drift population was subjected to six flush-crash cycles (Carson, 1971; Powell, 1978). One or two generations following each population crash (bottleneck), multiple-choice mating tests were performed on the 36 pairwise combinations of the eight drift lines and BASE. In each replicate test (8–32 replicates), we aspirated 24 three-day-old virgin flies, six of each line-sex combination, into a large plastic mating cell and recorded the cumulative number of matings at six-minute intervals for one hour. The tips of the wings of each line were clipped in half the replicates. Independent estimates of the mating frequencies (number mating/number observed) of BASE females and males were obtained from a parallel experiment (unpublished) in which inbred lines were derived by sib mating from the drift lines immediately after they were founded. The methods are described in greater detail by Ringo et al. (1985).

Testing the Hypotheses.—The three hypotheses predict qualitatively different relative numbers of matings in the 48 data sets of Ringo et al. (1985). To compare the observed pattern with these predictions, a two-way analysis of variance was performed on these data, using female and male type as factors. Then, cell means were ranked using Tukey's test (Sokal and Rohlf, 1981).

To visualize asymmetrical sexual isolation we plotted the percentage of homogamic matings by ancestral (BASE) females, $100x_{11}/(x_{11} + x_{21})$, against the percentage

of homogamic matings by derived (flush-crash populations) females, $100x_{22}/(x_{12} + x_{22})$, for each of the 48 mating tests. Some published data (Kaneshiro, 1976; Ohta, 1978; Powell, 1978; Dodd and Powell, 1985) were treated in the same way. Unfortunately, the first two of these experiments had unbalanced designs, so we had to adjust the data to correct for unequal sample sizes. To do this we used the observed mating frequencies and let the adjusted number of replicates per half experiment be half the total number of replicates. Thus, the values of the statistics used by Kaneshiro (1976) and Ohta (1978) were identical for the original and the adjusted data. Recall that measures of asymmetrical isolation may reflect only differences in male mating propensity.

Third, we used our data to test the predictions of the Kaneshiro and ethological hypotheses; recall that the Watanabe-Kawanishi hypothesis does not predict any P_{ij} for a mating experiment. The 48 mating tests were performed in six sets, each set being comprised of tests of the eight flush-crash populations with BASE after one of the six crashes. Since assortative mating was relatively stable from set to set (Ringo et al., 1985) we pooled the eight tests of the first set to estimate a , the parameter of assortative mating. The mating propensities were estimated from independent experiments, as noted above; thus all predictors were derived independently of the observed values in testing our ethological hypothesis. The remaining 40 data sets were used to test the model. In addition, the data of each set were pooled into a 2×2 table. The predicted and observed values of these five pooled data sets are presented; the analyses of the individual tests are summarized.

For both hypotheses, it is assumed that $Y_i = X_i + e_i$, where the Y_i 's and X_i 's are the observed and predicted values, respectively, and the e_i 's are independently, normally distributed error terms; goodness-of-fit of the data to this relationship was assessed by the SAS procedure NLIN (SAS Institute Inc., 1982 pp. 15-37). In this way, we analyzed x_{22} (matings between derived females and derived males) and x_{12} (matings between derived females and ancestral males). According to the ethological hypothesis, the Kaneshiro hypothesis yields biased predic-

TABLE 3. Mean number of matings in 48 tests using an ancestral population (BASE) and eight populations derived from it. Six females and six males per type per replicate were used, in 8-32 replicates. A total of 3,202 matings were observed, a mean of 66.8 matings per test. The derived populations underwent six "flush-crash" cycles—exponential growth followed by a bottleneck in size. Means with different letters (a, b, or c) are significantly different ($P < 0.05$) by Tukey's test. The observed pattern is that predicted by the ethological hypothesis, and does not conform to the patterns predicted by the Kaneshiro or Watanabe-Kawanishi hypotheses. The predicted patterns of means are qualitative: H = high, I = intermediate, and L = low. The pattern predicted by the ethological hypothesis is specific to this set of data; i.e., using another set of data, the hypothesis might predict another pattern.

Type of mating (female $\times$ male)	Observed	Pattern predicted by three hypotheses		
		Ethological	Kaneshiro	Watanabe
A $\times$ A	27.48 ^a	H	H	I
D $\times$ A	15.42 ^b	I	I	L
A $\times$ D	13.85 ^b	I	L	I
D $\times$ D	9.96 ^c	L	I	H

tions, and vice versa. The bias of each hypothesis predicted by the other was assessed using the fact that $\Sigma \chi_i / \sqrt{N}$ is approximately distributed as z , the standard normal variate.

For the Kaneshiro hypothesis, the values of x_{12} and x_{22} are predicted to be $f_{12}T/2$. For the ethological hypothesis, the p_i 's and P_j 's were estimated from the raw mean mating frequencies in independent mating tests using the nine lines: the mating frequencies of BASE were obtained from mating tests with a set of inbred lines, and those of the derived populations were obtained from mating tests among derived lines (tests not involving BASE). The P_{ij} 's were estimated from the marginals and a , as described above (see An Ethological Hypothesis of Mating Behavior); a worked example is given in the Appendix.

RESULTS

Analysis of variance revealed significant differences between female types ($F = 77.88$, $d.f. = 1,168$, $P < 0.0001$) and between male types ($F = 111.37$, $P < 0.0001$), as well as significant interaction, due to assortative mating ($F = 20.40$, $P = 0.0001$). The means are shown in Table 3, which compares the

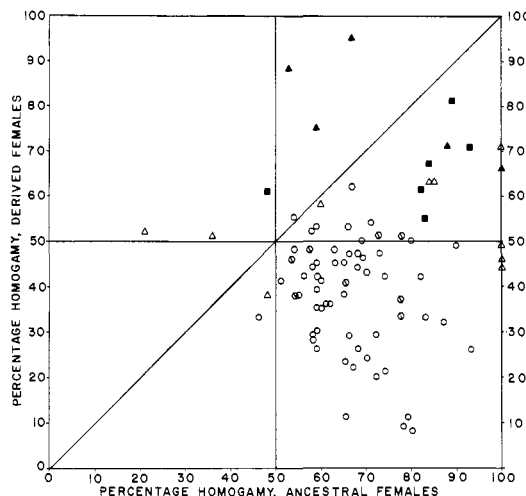


FIG. 1. Homogamic matings as a percentage of all matings by ancestral females vs. homogamic matings as a percentage of all matings by derived females. Circles: present study; octagons: Powell (1978); octagons with lines: Dodd and Powell (1985); squares: Kaneshiro (1976); triangles: Ohta (1978). Open symbols: intraspecific tests; solid symbols: interspecific tests.

observed pattern with that expected under the three models.

The percentage of homogamic matings by ancestral females is plotted against the percentage of homogamic matings by derived females in Figure 1 for 85 mating tests of five experiments (ours, and those of four published studies). The classification of populations as ancestral or derived in our experiment and in those of Powell (1978) and Dodd and Powell (1985) is unambiguous; we assumed Kaneshiro (1976) and Ohta (1978) to be correct in their designations of ancestral and derived. The Watanabe-Kawanishi hypothesis predicts homogamy in derived females and a greater tendency toward homogamy in derived than in ancestral females; i.e., it predicts that most points should lie in the triangle above the diagonal in the upper right-hand quadrant and in the upper left-hand quadrant; seven of the 85 observations lie in this trapezoid. The Kaneshiro hypothesis predicts homogamy in ancestral females and random mating in derived females; i.e., it predicts that most points should cluster around the line segment dividing the upper and lower right-hand quadrants. While such a tendency is

not apparent, there are obvious asymmetries: 1) ancestral females mated homogamically in nearly all cases, whereas derived females did not; and 2) homogamy was greater in ancestral females than in derived females, for nearly all cases. Most of the observations from Powell (1978) and Ringo et al. (1985) lie in the lower right-hand quadrant, which could be explained by the greater mating propensity of ancestral males in those studies. Populations which go through successive bottlenecks should be subject to inbreeding depression for male mating propensity (Spiess, 1970; Barton and Charlesworth, 1984) and perhaps for female mating propensity, as well (Ringo et al., 1986a).

The predictions of the ethological hypothesis and the Kaneshiro hypothesis are compared with observed values for the five data sets from our experiment in Table 4; in each set the data from eight tests are pooled. In the 40 individual mating tests, the predicted values of the x_{ij} based on the ethological hypothesis and predicted without knowledge of the total number mating, deviated significantly from observed numbers of matings in six cases. The predicted values of x_{12} and x_{22} based on the Kaneshiro hypothesis, and predicted with knowledge of the numbers of matings by derived females, deviated significantly from observed in 11 of the 40 tests. The predictions of the ethological hypothesis, made with knowledge of the numbers of matings by derived females but otherwise made from independently derived parameters (not from observed values in the 2×2 tables), deviated significantly from observed in 1 of the 40 tests. The sample sizes were so small for the derived females in these 40 mating tests (owing to the low mating propensities of derived females) that the tests of the hypotheses had very low power, and were therefore prone to Type II error. This was not true of the pooled data in Table 4, or of the overall assessment of bias. The Kaneshiro hypothesis consistently predicted higher than observed values for x_{22} and lower than observed values for x_{12} ($z = 7.59$, $P < 10^{-13}$), a bias predicted by the ethological hypothesis. The ethological hypothesis was not significantly biased ($z = 1.93$, $P = 0.062$). The predicted and observed

TABLE 4. Observed and predicted values for five sets of data from 40 mating tests of the 48 described in Table 1; the first set of eight mating tests was used to estimate α , the assortative mating parameter. Eight mating tests performed after each of the last five bottlenecks were pooled to yield the five data sets. Predicted values were obtained for the ethological hypothesis (E), the Kaneshiro hypothesis (K), and the ethological hypothesis given the number of matings by derived females (the same constraint used in the Kaneshiro hypothesis) (EC). For the first hypothesis, the Chi-square statistics are compared with Chi-square with 4 *df.*, since all four values are freely predicted without knowledge of the data. For the latter two hypotheses, the Chi-square statistics are compared with Chi-square with 1 *df.*

Data set	Value	Observed	E	χ^2	K	χ^2	EC	χ^2
2	x_{11}	213	199.0	10.61*	—	7.61*	—	0.35
	x_{12}	143	117.4		121.5		138.4	
	x_{21}	133	115.6		—		—	
	x_{22}	100	88.7		121.5		104.6	
3	x_{11}	233	206.7	4.02	—	4.13*	—	0.11
	x_{12}	124	120.2		109.0		126.4	
	x_{21}	116	115.2		—		—	
	x_{22}	94	87.1		109.0		91.6	
4	x_{11}	208	197.6	4.55	—	5.75*	—	0.49
	x_{12}	105	98.6		89.0		109.5	
	x_{21}	107	95.0		—		—	
	x_{22}	73	61.7		89.0		68.5	
5	x_{11}	215	213.5	3.83	—	13.31*	—	0.47
	x_{12}	132	112.8		105.5		127.1	
	x_{21}	103	108.4		—		—	
	x_{22}	79	74.4		105.5		83.9	
6	x_{11}	263	252.8	4.25	—	22.35*	—	2.35
	x_{12}	127	126.6		94.5		116.8	
	x_{21}	113	120.3		—		—	
	x_{22}	62	78.3		94.5		72.2	

* $P < 0.05$.

values of x_{22} (derived females with derived males) for the 40 mating tests are shown in Figure 2. The points for our hypothesis applied without knowledge of the number of matings (fig. 2A) tend to lie above this line because the total number of matings was consistently underestimated, but the fit of predicted points to the line is good ($r^2 = 0.75$). The points for the Kaneshiro hypothesis tend to lie below the line (Fig. 2B; $r^2 = 0.63$); an explanation for this is that derived males had lower mean mating propensities than ancestral males, a factor not taken into account by the Kaneshiro hypothesis. Our hypothesis applied with knowledge of the number of matings by derived females (as in the Kaneshiro hypothesis) improved the goodness-of-fit to the line (Fig. 2C; $r^2 = 0.85$).

DISCUSSION

We observed asymmetries in mating tests between ancestral and derived "flush-crash" populations and in reanalyzed data from

studies by Kaneshiro (1976) and Ohta (1978); homogamic matings were more frequent in ancestral females. Our analysis of data from experiments using *D. simulans* and *D. pseudoobscura* rejects the hypothesis of Watanabe and Kawanishi (1979) for populations having the properties assumed in the model of Kaneshiro (1976); i.e., the derived populations were subjected to flush-crash cycles (Carson, 1971) after they had budded off the large, stable ancestral population, and there was no gene flow between populations (Giddings and Templeton, 1983).

A stronger asymmetry was apparent in our experiment: ancestral females and males had higher mean mating propensities than did derived females and males. There is some evidence to suggest that inbreeding depression is the cause (Ringo et al., 1986a). Thus, the asymmetry due to female-homogamy interaction in individual mating tests does not imply asymmetrical discrimination. Our hypothesis, which takes mating

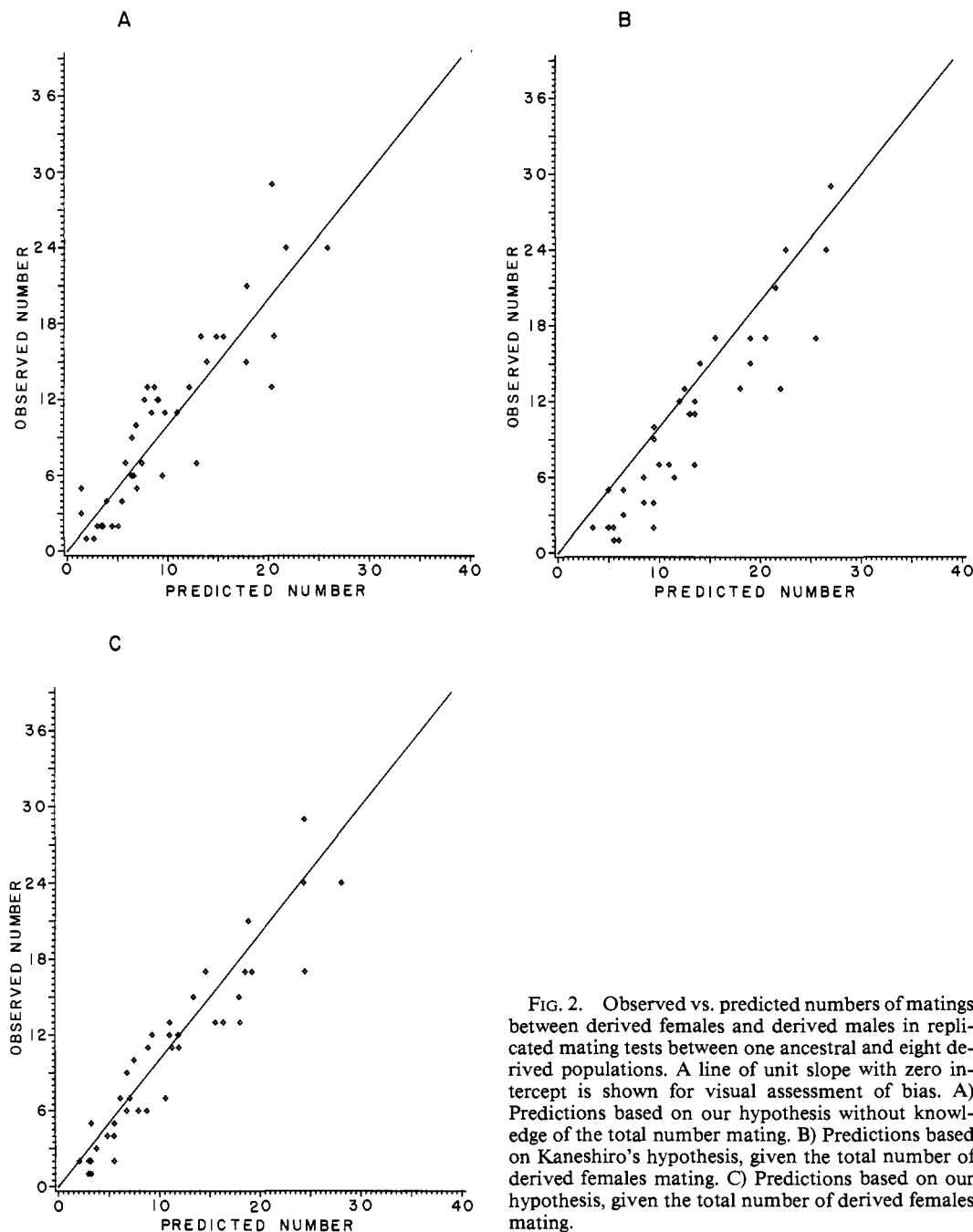


FIG. 2. Observed vs. predicted numbers of matings between derived females and derived males in replicated mating tests between one ancestral and eight derived populations. A line of unit slope with zero intercept is shown for visual assessment of bias. A) Predictions based on our hypothesis without knowledge of the total number mating. B) Predictions based on Kaneshiro's hypothesis, given the total number of derived females mating. C) Predictions based on our hypothesis, given the total number of derived females mating.

propensities into account, predicts the results of our experiment more accurately than does the Kaneshiro hypothesis. The predictions of our hypothesis are unbiased, which is the strongest piece of evidence against asymmetrical mate discrimination in this

experiment. Our hypothesis has the added advantages of predicting all the mating frequencies in an experiment, and of being applicable to any two populations or species.

Our ethological hypothesis does not relate mating frequencies to the "direction of evo-

lution." However, the concept of ancestral and derived populations or species has limited value. Under the conditions defined by Giddings and Templeton (1983), the ancestral population changes much less than the derived populations after the split between them, but it is not clear that even Hawaiian *Drosophila* species conform to these conditions. The concepts of ancestral and derived apply to species only if evolution proceeds much more rapidly in one lineage than in another. There is no evidence that this occurs in the long run for the entire genome, even though some traits may be ancestral and some may be derived. Thus, for example, we are reluctant to subscribe either to the view that *Homo sapiens* is ancestral to *Pan troglodytes* or to the view that *Pan* is ancestral to *Homo*. On the contrary, we suppose that both man and chimp have undergone substantial amounts of evolutionary change since the two lineages split, and that there is no vector between the species defining "the direction of evolution."

Our ethological hypothesis has good predictive power primarily because the populations tested have stable mating propensities and a low degree of assortative mating. The mating frequencies of both sexes of a given type varied little from test to test, regardless of the second type used in the mating test; e.g., about 60% of the BASE females mated in one hour, consistently over a three-year period, in mating tests using 27 different populations. The hypothesis will not be effective in predicting the results of interspecific mating tests, because positive assortative mating (high value of a) dominates such experiments, and the degree of assortative mating between two species cannot be accurately predicted from independent experiments. Nonetheless, it will be useful in assessing the asymmetry of sexual isolation.

To test the three competing hypotheses requires more than the observation of mating frequencies: the behavioral causes of variation in mating frequencies must be elucidated, as well. Kaneshiro's hypothesis predicts that derived males will have lost courtship elements. A study of the behavioral elements of courtship in BASE and two of our derived populations having very low

mating propensities showed that no courtship elements had been lost by derived males (Ringo et al., 1986a). Males of one derived population courted less than BASE males, whereas males of the other derived population did not. The proportion of courtship devoted to the two wing displays, scissoring and vibration, did not vary significantly among populations. The three types of male did not differ significantly in surface hydrocarbons thought to play a pheromonal role (J.-M. Jallon, pers. comm.). The ethological model predicts that males of the three populations will differ in mating success in "no choice" mating tests and that they will differ in measures of mating rate. Both predictions are borne out (Ringo et al., 1986a; Ringo et al., 1986b).

ACKNOWLEDGMENTS

The data were collected with support from NSF Grant DEB 7903866. The Computing Center of the University of Maine provided assistance. We thank William Glanz for his helpful advice and comments throughout the study, and Phyllis Brooks and Eddie Meisner for help in preparing the manuscript. Brian Charlesworth and Eliot Spiess provided careful, thoughtful reviews.

LITERATURE CITED

- BARTON, N. H., AND B. CHARLESWORTH. 1984. Founder effects, genetic revolutions, and speciation. *Ann. Rev. Ecol. Syst.* 15:133-164.
- BASTOCK, M. 1967. Courtship. An Ethological Study. Aldine, Chicago, IL.
- BASTOCK, M., AND A. MANNING. 1955. The courtship of *Drosophila melanogaster*. *Behaviour* 8:85-111.
- BATEMAN, A. J. 1949. Analysis of data on sexual isolation. *Evolution* 3:174-177.
- BISHOP, Y. M. M., S. E. FIENBERG, AND P. M. HOLLAND. 1975. Discrete Multivariate Analysis: Theory and Practice. MIT Press, Cambridge, MA.
- CARSON, H. L. 1971. Speciation and the founder principle. *Stadler Genet. Symp.* 3:51-70.
- DOBZHANSKY, TH. 1944. Experiments on sexual isolation in *Drosophila*. III. Geographic strains of *D. sturtevantii*. *Proc. Nat. Acad. Sci. USA* 30:335-339.
- DOBZHANSKY, TH., AND G. STREISINGER. 1944. Experiments on sexual isolation in *Drosophila*. II. Geographic strains of *Drosophila prosaltans*. *Proc. Nat. Acad. Sci. USA* 30:340-345.
- DODD, D. M. B., AND J. R. POWELL. 1985. Founder-flush speciation: An update of experimental results with *Drosophila*. *Evolution* 39:1388-1392.
- EHRMAN, L., AND C. PETIT. 1968. Genotype frequency and mating success in the *willistoni* species group of *Drosophila*. *Evolution* 22:649-658.
- GIDDINGS, L. V., AND A. R. TEMPLETON. 1983. Be-

- havioral phylogenies and the direction of evolution. *Science* 220:372-378.
- GOUX, J. M., AND D. ANXOLABEHÈRE. 1980. The measurement of sexual isolation and selection: A critique. *Heredity* 45:255-262.
- KANESHIRO, K. Y. 1976. Ethological isolation and phylogeny in the *planitibia* subgroup of Hawaiian *Drosophila*. *Evolution* 30:740-745.
- . 1983. Sexual selection and direction of evolution in the biosystematics of Hawaiian *Drosophilidae*. *Ann. Rev. Entomol.* 28:161-178.
- LEVENE, H. 1949. A new measure of sexual isolation. *Evolution* 3:315-321.
- MAYR, E. 1963. *Animal Species and Evolution*. Belknap, Cambridge, MA.
- OHTA, A. T. 1978. Ethological isolation and phylogeny in the *grimshawi* species complex of Hawaiian *Drosophila*. *Evolution* 32:485-492.
- PATTERSON, J. T., AND L. W. McDANALD, AND W. S. STONE. 1947. Sexual isolation between members of the *virilis* group of species. *Stud. Genet. Univ. Texas Publ.* 4720:7-31.
- PATTERSON, J. T., AND W. S. STONE. 1952. *Evolution in the Genus Drosophila*. Macmillan, N.Y.
- PETIT, C., D. KITAGAWA, AND T. TAKAMURA. 1976. Mating systems between Japanese and French geographic strains of *Drosophila melanogaster*. *Jap. J. Genet.* 51:99-108.
- POWELL, J. R. 1978. The founder-flush speciation theory: An experimental approach. *Evolution* 32:465-474.
- RINGO, J. M., K. M. BARTON, AND H. B. DOWSE. 1986a. The effect of genetic drift on mating propensity, courtship behaviour, and postmating fitness in *Drosophila simulans*. *Behaviour. In press*.
- RINGO, J. M., H. B. DOWSE, AND K. M. BARTON. 1986b. Courtship behavior and the kinetics of mating in three lines of *Drosophila simulans* and their hybrids. *Behav. Genet. In press*.
- RINGO, J. M., D. WOOD, R. ROCKWELL, AND H. B. DOWSE. 1985. An experiment testing two hypotheses of speciation. *Amer. Natur.* 126:642-661.
- SAS INSTITUTE, INC. 1982. *User's Guide: Statistics*. SAS Inst. Inc., Cary, NC.
- SCHAFER, H. E. 1968. A measure of discrimination in mating. *Evolution* 22:125-129.
- SCHILCHER, F. VON, AND M. DOW. 1977. Courtship behavior in *Drosophila*. Sexual isolation or sexual selection? *Z. Tierpsych.* 43:304-310.
- SOKAL, R. R., AND F. J. ROHLF. 1981. *Biometry*. Freeman, San Francisco, CA.
- SPIESS, E. B. 1970. Mating propensity and its genetic basis in *Drosophila*, pp. 315-375. In M. K. Hecht and W. C. Steere (eds.), *Essays in Evolution and Genetics in Honor of Theodosius Dobzhansky*. Appleton-Century-Crofts, N.Y.
- SPIETH, H. T. 1949. Sexual behavior and isolation in *Drosophila*. II. The interspecific mating behavior of species of the *willistoni* group. *Evolution* 3:67-81.
- . 1951. Mating behaviour and sexual isolation in the *Drosophila virilis* species group. *Behaviour* 3:105-145.
- . 1968. The evolutionary implications of sexual behavior in *Drosophila*. *Evol. Biol.* 2:157-191.
- . 1974. Courtship behavior in *Drosophila*. *Ann. Rev. Entomol.* 19:385-405.
- SPIETH, H. T., AND J. M. RINGO. 1983. Mating behavior and sexual isolation in *Drosophila*, pp. 223-284. In M. Ashburner, H. L. Carson, and J. N. Thompson, Jr. (eds.), *The Genetics and Biology of Drosophila*. Academic Press, N.Y.
- STALKER, H. B. 1942. Sexual isolation studies in the species complex *Drosophila virilis*. *Genetics* 27:238-257.
- TAYLOR, C. E. 1975. Differences in mating propensities: Some models for examining the genetic consequences. *Behav. Genet.* 5:381-393.
- WATANABE, T. K., AND M. KAWANISHI. 1979. Mating preference and the direction of evolution in *Drosophila*. *Science* 205:905-907.
- . 1981. Asymmetrical mating success and the phylogeny of *Drosophila*. *Zool. Mag.* 90:317-324.
- WOOD, D., AND J. M. RINGO. 1980. Mating discrimination in *Drosophila melanogaster*, *D. simulans*, and their hybrids. *Evolution* 34:320-329.

Corresponding Editor: J. B. Spofford

APPENDIX

Predicting Numbers of Matings from the Models: A Worked Example.—In this example, the fractions of ancestral and derived females mating in separate experiments, independent of the one being analyzed below (see Materials and Methods), were:

Ancestral females	0.6097
Derived females	0.1597
Ancestral males	0.6138
Derived males	0.2139.

Notice that the female and male mating frequencies are unequal. This is because ancestral and derived frequencies were obtained from different experiments (see Methods and Materials). In the mating test, there were 12 replicates in which six females and six males of each type were used; the observed numbers of matings were:

x_{11}	33
x_{12}	5
x_{21}	17
x_{22}	2.

Our hypothesis.—First, estimate the total number mating, $T = (0.6097 + 0.1597)(6)(12) = 55.4$; this is not very different from the observed number, $T = 57$.

Second, estimate the relative mating propensities:

$$\begin{aligned}
 P_1 &= (0.6097)/(0.6097 + 0.1597) = 0.7924 \\
 P_2 &= 1 - P_1 = 0.2076 \\
 P_1 &= (0.6138)/(0.6138 + 0.2139) = 0.7416 \\
 P_2 &= 1 - P_1 = 0.2584.
 \end{aligned}$$

Third, estimate numbers of matings assuming no assortative mating:

$$\begin{aligned}
 X &= (0.7924)(0.7416)(55.4) = 32.56 \\
 Y &= (0.2076)(0.7416)(55.4) = 8.53 \\
 Z &= (0.7924)(0.2584)(55.4) = 11.34 \\
 W &= (0.2076)(0.2584)(55.4) = 2.97 \\
 T &= 55.40.
 \end{aligned}$$

Fourth, correct for assortative mating. To adjust the values X , Y , Z , and W , we need to find K such that $(X + K)(W + K)/(Y - K)(Z - K) = a$. Solving the quadratic equation,

$$(a - 1)K^2 - (aY + aZ + X + W)K + aYZ - XW = 0$$

for $a = 1.3$ (the value determined in an independent experiment), we get $K = 0.47$. K is related simply to k , mentioned in the section on our ethological hypothesis: $K = kT$.

Finally, we find the x_{ij} , rounding to the nearest tenth:

$$x_{11} = 32.56 + 0.47 = 33.0$$

$$x_{12} = 8.53 - 0.47 = 8.1$$

$$x_{21} = 11.34 - 0.47 = 10.9$$

$$x_{22} = 2.97 + 0.47 = 3.4.$$

Testing goodness-of-fit with chi-square, $X^2 = 4.934$. There are 4 *df*, since all four expected values are estimated freely, without using the observed values. X^2

$< \chi^2(4, 0.05) = 9.488$ and we fail to reject the null hypothesis.

Using the total number of matings by derived females ($5 + 2 = 7$) to predict x_{12} and x_{22} , and calculating X^2 for goodness-of-fit,

$$x_{12} = (8.1)(7)/(11.5) = 4.9$$

$$x_{22} = (3.4)(7)/(11.5) = 2.1$$

$X^2 = 0.007$. There is only 1 *df*, since the observed column total was used. $X^2 < \chi^2(1, 0.05) = 3.841$, and we fail to reject the null hypothesis. The power of this chi-square test is very low, but the predicted value of x_{22} is only 5% larger than the observed value.

The Kaneshiro hypothesis.—This hypothesis predicts that $x_{12} = x_{22} = x_2/2 = 7/2 = 3.5$; $X^2 = 0.823$. It also predicts that $x_{11} > x_{21}$; $X^2 = 5.12$. Both predictions are borne out, since we reject the null hypothesis with ancestral females and accept it with derived females. Again, the power of the chi-square test is very low. In this case, the predicted value of x_{22} is 75% larger than the observed value.